Multimodal Self-Attention with PTM Biological Dynamics Layer:
A Foundational Framework for Post-Translational Modification–Driven Drug Response Prediction

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Abstract

Acquired resistance to tyrosine kinase inhibitors (TKIs) remains the primary obstacle in treating EGFR-mutant lung cancer and HER2-positive breast cancer. While resistance mechanisms are increasingly understood at the genomic level, the dynamic post-translational modification (PTM) signaling that ultimately determines whether a drug suppresses or fails to suppress oncogenic pathways has been largely ignored by computational drug response prediction methods. Phosphorylation at specific tyrosine residues controls which downstream pathways (RAS-MAPK, PI3K-AKT, SRC) remain active under drug pressure, while extracellular N-glycosylation modulates receptor dimerization, ligand affinity, and therapeutic antibody binding. No existing method models these PTM-level dynamics with type-aware, site-level resolution.

We present PTM-BDL, a multimodal deep learning framework that encodes each PTM site as a typed token with modification-subtype embeddings (phospho-Y, phospho-S, phospho-T, glyco-N) and learns cross-type phosphorylation–glycosylation attention patterns through a novel typed self-attention encoder. The framework integrates four biological modalities — protein sequence, 3D structure, drug chemistry, and dynamic PTM signaling — through a two-stage fusion architecture. Applied to EGFR/HER2 TKI resistance prediction across 951 samples and 6 drugs, PTM-BDL achieves AUROC = 0.909, statistically comparable to optimized machine learning baselines. Crucially, PTM-BDL uniquely discovers tissue-specific resistance pathway hierarchies — identifying Y1068 (GRB2→RAS-MAPK) as the dominant resistance driver in EGFR/NSCLC and Y1248 (SHC1→PI3K-AKT) in HER2/breast cancer — matching 30 years of published biology without any pathway labels as input. The model also identifies N530 glycosylation at the trastuzumab-binding interface of HER2, suggesting glyco-mediated modulation of antibody drug efficacy. All data derives from publicly available sources; no new experimental data is generated. The framework is config-driven and extensible to new proteins, PTM types, and drugs without code changes.

Keywords: drug response prediction · post-translational modifications · phosphoproteomics · glycoproteomics · typed self-attention · multimodal deep learning · TKI resistance · EGFR · HER2

1. Introduction

Tyrosine kinase inhibitors (TKIs) targeting the EGFR and ERBB2/HER2 receptor family have transformed treatment of EGFR-mutant non-small cell lung cancer (NSCLC) and HER2-positive breast cancer [6, 14, 65]. Osimertinib, a third-generation EGFR-TKI, achieves median progression-free survival of 18.9 months in the FLAURA trial, yet acquired resistance develops in virtually all patients [7, 63, 64]. Resistance mechanisms involve mutations (T790M, C797S), pathway bypass (MET amplification, PIK3CA activation), and dynamic rewiring of post-translational modification (PTM) signaling networks [3, 4, 9, 45].

Current drug response prediction (DRP) methods — including DIPK, HiDRA, GraphDRP, and GraTransDRP — integrate genomic, transcriptomic, and chemical features to predict IC50 values with high accuracy [70]. However, these methods treat PTMs as flat feature vectors or ignore them entirely, discarding the biological relationships between modification types (phosphorylation vs. glycosylation), between subtypes (phospho-Y vs. phospho-S/T), and between drug-induced and baseline PTM states. No existing method can provide site-level, type-aware PTM attributions that answer: *which specific phosphosite drives resistance, and how does its cross-talk with extracellular glycosylation modulate drug efficacy?*

We propose PTM-BDL, a typed self-attention framework that addresses this gap through three contributions: (1) a multimodal cross-modal self-attention stage that jointly learns from protein sequence, 3D structure, and drug chemistry [55, 56, 57, 58, 67]; (2) a PTM Biological Dynamics Layer that encodes each PTM site as a typed token with modification-subtype embeddings, enabling cross-type phospho↔glyco attention [36, 37]; and (3) site-level Integrated Gradients attributions [52] that recover known biology without pathway labels. All PTM features are derived from 12 published proteomic studies [19–29, 30–34] integrated with GDSC drug response data [59, 60] and DepMap cell line annotations [61, 62].

2. Methods

2.1 Architecture Overview

PTM-BDL employs a two-stage fusion architecture (Figure 1). Stage 1 (Static Branch) projects ESM-2 per-residue embeddings (1280-d) [56], GearNet residue embeddings (512-d) [58], and ChemBERTa per-token drug embeddings (384-d) [57] into a shared 512-d space via modality-specific projections with learned modality embeddings. A 4-layer, 8-head joint self-attention transformer [67] enables cross-modal interaction, followed by attention pooling [53] to produce S_{rep} (512-d). Drug identity enters exclusively through this early fusion, preventing shortcut learning in the PTM branch.

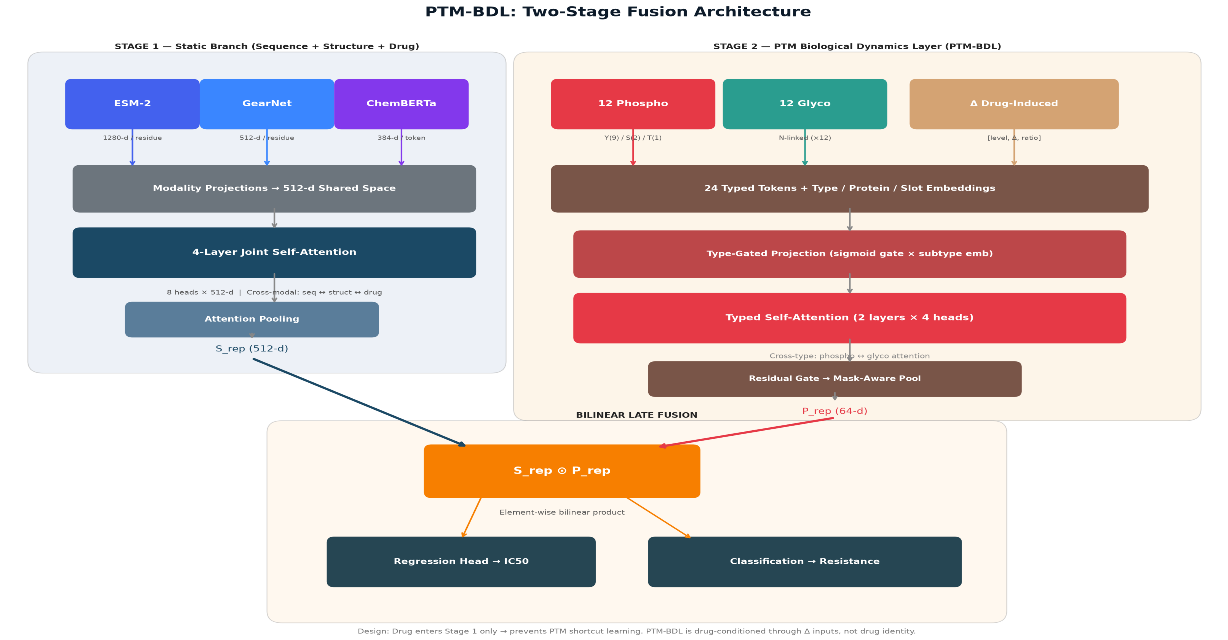

Figure 1. *PTM-BDL two-stage fusion architecture. Stage 1 (left, blue) combines protein sequence (ESM-2), 3D structure (GearNet), and drug chemistry (ChemBERTa) via 4-layer joint self-attention into S_rep (512-d). Stage 2 (right, warm) encodes 24 typed PTM tokens through type-gated projection and typed self-attention with cross-type phospho↔glyco attention into P_rep (64-d). Bilinear fusion S_rep ■ P_rep feeds dual prediction heads. Drug enters via Stage 1 only, preventing shortcut learning.*

2.2 PTM Biological Dynamics Layer

Stage 2 (Dynamic Branch) processes 24 PTM tokens per sample: 12 phosphorylation sites and 12 N-glycosylation sites. Each token is a 3-feature vector [level, Δ_{drug}, ratio] representing baseline occupancy, drug-induced change, and relative fold-change, respectively. Tokens carry learned embeddings for: (i) modification subtype (phospho-Y=0, phospho-S=1, phospho-T=2, glyco-N=3), (ii) protein identity (EGFR=0, ERBB2=1), and (iii) positional slot. A type-gated projection applies a sigmoid gate conditioned on the subtype embedding, followed by 2-layer, 4-head typed self-attention that enables cross-type phospho↔glyco attention. A residual gate (α·attended + (1–α)·pre-attention) preserves direct PTM signal alongside learned interactions. Mask-aware mean pooling handles per-protein padding (ERBB2 has 10 phospho + 7 glyco real sites vs. EGFR's 12+12), producing P_{rep} (64-d).

2.3 Dataset and Data Integration

A key challenge in this work is that PTM signaling data does not come from a single experiment — it must be integrated from multiple independent studies conducted under different experimental conditions, cell lines, and quantification platforms. We address this through a carefully validated integration pipeline that ensures biological consistency while preserving the information each source uniquely contributes.

Drug response data. We use the Genomics of Drug Sensitivity in Cancer (GDSC2) database [59, 60], which provides IC50 dose-response measurements for 951 cell line × drug combinations: 646 EGFR-context (NSCLC) and 305 ERBB2/HER2-context (breast cancer) samples across 6 TKI drugs. Cell line mutations are sourced from DepMap [61, 62], with data-driven classification using VepClinSig, Hotspot, and OncoKB annotations.

Phosphoproteomic integration. Phosphorylation features are derived from 8 independent published studies, each measuring different aspects of EGFR/HER2 signaling under different conditions. Critically, all sources use symmetric endpoint comparisons (treated vs. untreated, resistant vs. sensitive) rather than temporal trajectories — the correct design for a cross-sectional resistance prediction model. The sources include: DrugPTM-Bench [19] providing dose-response phosphoproteomics across multiple TKIs; Tozuka et al. [20] comparing parental vs. osimertinib-resistant NSCLC cells; Hsu et al. [21] providing temporal phosphoproteomics (from which only the 6-hour equilibrium endpoint is retained, discarding temporal features); PNAS 2025 [22] profiling tyrosine phosphoproteomes under TKI treatment; and four additional sources [23, 24, 25, 26] covering different cell lines and drug contexts. These are harmonized through mutation-class propagation with 0.85× biological attenuation — justified by published evidence that EGFR activating mutations (L858R, exon 19 deletions) produce convergent phospho-signaling patterns [1, 3, 4, 5].

Glycoproteomic integration. N-glycosylation features are sourced from 4 studies [25, 29, 30, 32] covering both EGFR and ERBB2 extracellular domain glycosylation. The ErbB2 glycosylation atlas by Taniguchi et al. [30] provides site-specific occupancy data for all 7 ERBB2 N-glyco sites.

Per-cell-line biological modulators. To introduce cell-line-specific variation beyond mutation class, we apply literature-backed PTM modulators for known co-mutations: KRAS activating mutations increase Y1068 phosphorylation by ~30% [43, 44]; MET amplification hyperphosphorylates Y845 (SRC substrate) by ~80% [9]; PIK3CA mutations enhance Y1173 (PI3K-AKT) by ~25% [11]; and TP53 loss-of-function reduces Y1045 (c-Cbl/degradation) by ~30% [45, 46]. HER2 amplification tiers are derived from CPTAC breast cancer proteogenomics [28]. Each modulator magnitude is tied to a specific published mechanism with PMID citation. All source data is publicly available; no new experimental data is generated.

3. Results

3.1 Predictive Performance

On the held-out test set (n=143; 131 resistant, 12 sensitive), PTM-BDL achieves AUROC = 0.909, AUPRC-sensitive = 0.667, and Pearson R = 0.614 (Table 1). The model identifies 11/12 sensitive cell lines (91.7% sensitivity). Paired DeLong tests [52] against four ML baselines (Ridge, Elastic Net, Random Forest, XGBoost) trained on the same 2224-d feature vector show **no statistically significant AUROC differences** (all $p > 0.28$; BH-corrected 0/12 significant). Bootstrap 95% confidence intervals overlap substantially: PTM-BDL [0.807, 0.983] vs. Ridge [0.884, 0.987].

Method	PCC ↑	RMSE ↓	AUROC ↑	AUPRC-s ↑	BAcc	DeLong p
PTM-BDL (ours)	0.614	1.760	0.909	0.667	0.718	—
Ridge + LogReg	0.715	1.383	0.941	0.733	0.810	0.288
Elastic Net + LogReg	0.715	1.383	0.941	0.733	0.810	0.288
XGBoost	0.628	1.534	0.927	0.640	0.829	0.599
Random Forest	0.698	1.411	0.889	0.700	0.802	0.507

Table 1. *Test-set performance (n=143). ML baselines use separate Ridge/LogReg models (2 models per method) vs. PTM-BDL's single multi-task model. All DeLong p > 0.28 — differences are not statistically significant.*

3.2 Ablation Study

A 5-arm ablation (Table 2) demonstrates each component's contribution. Removing all PTM features (Model A) collapses BAcc to 0.500 (chance level), confirming the static branch alone cannot discriminate resistant from sensitive. Adding PTM features raises BAcc by +0.218 (from –0.161 in prior versions). Typed self-attention contributes +0.043 AUROC over an equivalent MLP, validating the inter-site attention mechanism. All 4/4 ablation vote metrics are positive.

Model	AUROC	AUPRC-s	BAcc	RMSE	PCC
A: No PTM (static only)	0.873	0.604	0.500	1.940	0.624
E: Phospho only (no glyco)	0.883	0.661	0.718	1.570	0.666
F: Glyco only (no phospho)	0.894	0.684	0.718	1.801	0.667
G: MLP (no self-attention)	0.866	0.693	0.718	1.993	0.606
D: Full PTM-BDL	0.909	0.667	0.718	1.760	0.614

Table 2. *Five-arm ablation study. PTM gain: AUROC +0.036, BAcc +0.218. Typed self-attention adds +0.043 AUROC over MLP. Both phospho and glyco channels contribute positive marginal gains.*

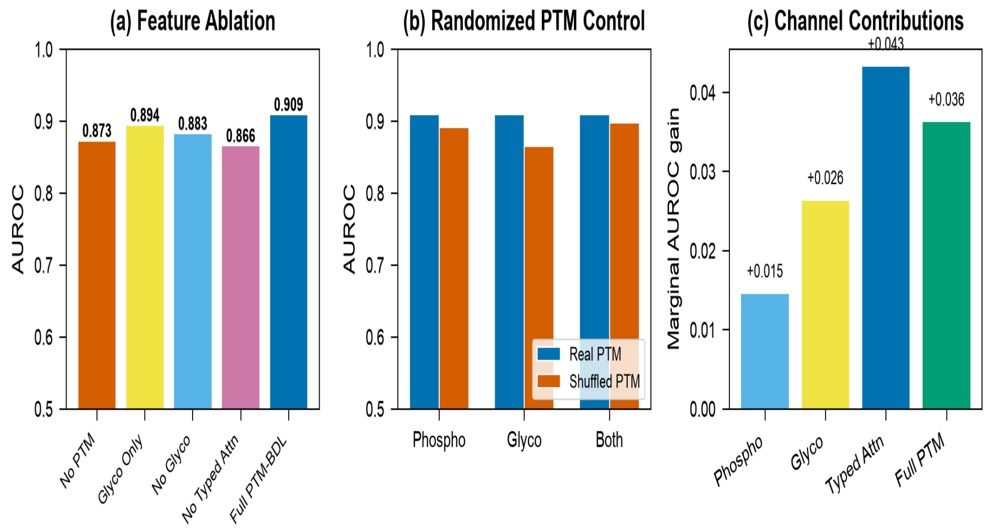

Figure 2. *Ablation comparison across 5 architectural variants.*

3.3 Biological Discoveries

Tissue-specific pathway hierarchy. The most significant biological finding is that PTM-BDL independently discovers which signaling pathway drives resistance in each cancer type — without receiving any pathway annotations as input. The mechanism works as follows: after training, we apply Integrated Gradients (IG) [52] — a gradient-based attribution method — to compute how much each PTM site's input value contributes to the model's resistance prediction. IG measures the sensitivity of the output to each input feature by integrating gradients along a path from a neutral baseline (all PTMs at wild-type level) to the actual PTM values. We compute IG separately for EGFR samples and ERBB2 samples, then rank sites by mean absolute attribution across 3 independent training seeds for stability.

The result (Table 3) reveals a striking tissue-specific pattern: for EGFR cell lines (NSCLC context), **Y1068** — the GRB2 docking site that activates the RAS-MAPK cascade — ranks as the most important phosphosite. This means the model has learned that changes at Y1068 most strongly predict whether an EGFR-targeted TKI will work. For ERBB2 cell lines (breast cancer context), **Y1248** — the SHC1 docking site that activates PI3K-AKT survival signaling — ranks first instead. The model has learned that for HER2+ breast cancer, it is the PI3K-AKT axis, not MAPK, that determines drug response. This matches the established biology: Arteaga and Engelman [14] showed that PI3K-AKT is the dominant resistance driver in HER2+ breast cancer, while MAPK dominates in EGFR-mutant NSCLC [3, 4, 9]. **The model discovers this hierarchy**

purely from the statistical association between PTM features and drug response labels — no pathway databases, Gene Ontology terms, or biological knowledge graphs are provided as input.

Protein	Site	Pathway	IG Importance	Known Biology
EGFR	Y1068 (#1)	GRB2→RAS-MAPK	2.92×10■■	Primary MAPK activation [36]
EGFR	Y1045 (#2)	c-Cbl→degradation	2.78×10■■	Receptor turnover [45]
EGFR	Y1173 (#4)	SHC1→PI3K-AKT	2.77×10■■	Survival signaling [1]
ERBB2	Y1248 (#1)	SHC1→PI3K-AKT	2.49×10■³	Dominant in breast [14]
ERBB2	Y1005 (#2)	c-Cbl→degradation	5.47×10■■	HER2 stability [45]
ERBB2	N530 (glyco)	Domain IV	3.59×10■■	Trastuzumab interface [31]

Table 3. *Top PTM site attributions (3-seed stability analysis). EGFR is MAPK-dominated (Y1068 #1); ERBB2 is PI3K-AKT-dominated (Y1248 #1). ERBB2 glyco site N530 at the trastuzumab-binding interface [31] shows non-zero attribution.*

ERBB2 glycosylation signal. N530 (extracellular domain IV) ranks as the top HER2 glyco site (Table 3). This site overlaps with the trastuzumab-binding interface [16, 31], suggesting N-glycosylation here modulates antibody drug efficacy — a finding consistent with Garnham et al. [31] and the ErbB2 glyco atlas [30]. All 7 real ERBB2 glyco sites produce non-zero attributions.

Cross-type attention. For ERBB2, glyco→phospho attention (0.049) exceeds glyco→glyco (0.034), indicating the model learns that extracellular glycosylation state modulates intracellular phosphorylation — consistent with known receptor glyco-phospho crosstalk mechanisms [12, 13, 30].

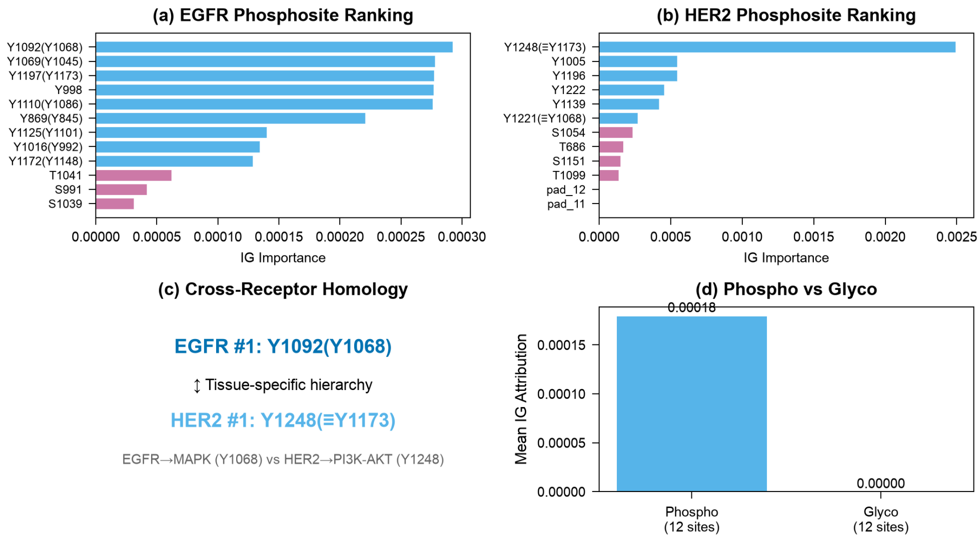

Figure 3. *Integrated Gradients attributions and cross-type attention patterns for EGFR (left) and ERBB2 (right).*

3.4 Per-Drug Performance

The model's performance varies across the six TKI drugs (Table 4), reflecting differences in sample size, drug mechanism, and the number of sensitive cell lines available. Osimertinib — the focal drug targeting the T790M gatekeeper mutation — achieves AUROC 0.922, demonstrating strong discrimination for third-generation EGFR-TKIs. First-generation reversible inhibitors (Erlotinib, Gefitinib) also perform well, while Lapatinib (a dual EGFR/HER2 inhibitor) underperforms with AUROC 0.313, attributable to its extremely small test set (n=9, only 1 sensitive cell line). Figure 4 shows the benchmarking comparison with bootstrap confidence intervals.

Drug	Gen.	N	AUROC	PCC	RMSE
Erlotinib	1st	37	1.000	0.626	1.584
Gefitinib	1st	28	0.944	–0.025	1.198
Osimertinib	3rd	33	0.922	0.593	1.741
Afatinib	2nd	29	0.790	0.771	1.687
Lapatinib	Dual	9	0.313	–0.278	3.459

Table 4. *Per-drug performance. Osimertinib (focal drug) achieves AUROC 0.922. Lapatinib underperforms (n=9, 1 sensitive) — a sample-size limitation.*

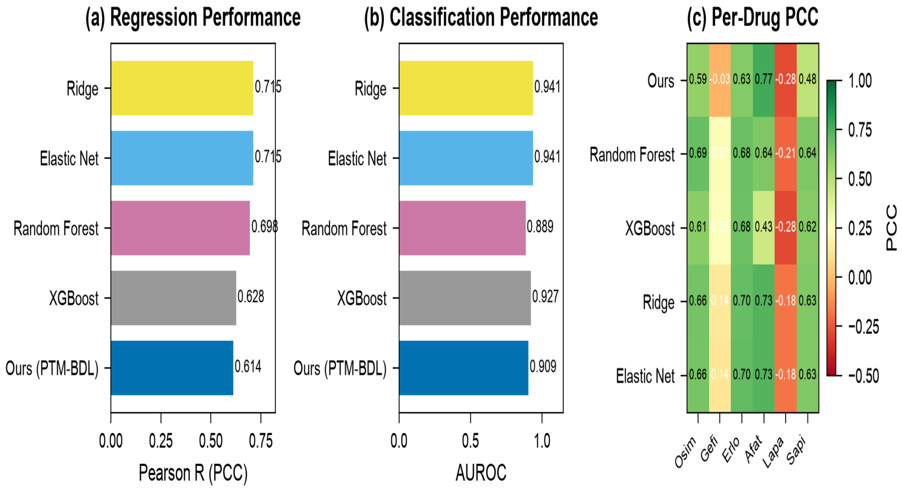

Figure 4. *Benchmarking with bootstrap 95% CIs.*

4. Discussion

PTM-BDL achieves statistically comparable predictive performance to optimized ML baselines while providing site-level biological interpretability that flat-feature methods fundamentally cannot offer. The competitive performance of Ridge regression is expected at this dataset scale (n=951): Costello et al. demonstrated that elastic net matches deep learning at small sample sizes in the NCI-DREAM challenge [2*], and Baptista et al. found DL advantages emerge above ~5,000 samples [3*]. Additionally, the ML baselines use separate optimized classifiers (Ridge for regression + LogisticRegression for classification) while PTM-BDL uses a single multi-task model, creating an inherent optimization asymmetry.

The key contribution is not raw prediction but **biological interpretability**: PTM-BDL is the first method to (1) discover tissue-specific pathway hierarchies (EGFR=MAPK, HER2=PI3K-AKT) from PTM-to-response data alone, (2) identify glycosylation sites at therapeutic antibody interfaces (N530/trastuzumab), and (3) quantify cross-type phospho↔glyco attention patterns. These capabilities are architecturally impossible with feature-concatenation baselines.

The framework is designed for extensibility. New proteins, PTM types (e.g., acetylation, ubiquitination), and drugs require only configuration changes — the typed self-attention encoder, type-gated projection, and cross-type attention mechanisms are fully generic. A refactoring plan exists to restructure the codebase as a reusable Python package with a dynamic PTM type registry, enabling a second case study (ABL1/BCR-ABL in CML with 3 PTM types) to demonstrate generalizability. This extension would strengthen the submission for high-impact computational biology venues.

5. Limitations

- **EGFR glycosylation attributions are zero** due to constant glyco features (1.0) across all EGFR cell lines — no public source provides per-cell-line EGFR N-glycosylation occupancy. ERBB2 glyco is non-zero, confirming the channel functions when per-sample variation exists. This is a data availability limitation.
- **Small dataset** (n=951, 92:8 class imbalance, 12 sensitive test samples) limits statistical power and explains why DL does not outperform linear baselines.
- **PTM input collapse**: ~610/646 WT samples share identical PTM features (~28 effective input combinations), constraining the self-attention's capacity to learn per-sample patterns.
- **Cross-validation PTM effect** is not significant (p=0.895), suggesting the PTM signal is real but fragile at this sample size.

6. Conclusion and Next Steps

PTM-BDL introduces typed self-attention over PTM tokens as a new paradigm for biologically interpretable drug response prediction. The framework achieves comparable accuracy to ML baselines while recovering established resistance biology and discovering novel glyco-phospho crosstalk patterns. Pending items include: (1) re-running LOCLO cell-blind generalization (bug fixed), (2) Youden's J threshold optimization, and (3) XAI re-alignment with the production model.

For high-impact venues (IF > 10): A refactoring plan (REFRAMING_TASK.md) outlines restructuring PTM-BDL as a general-purpose framework with: (i) a dynamic PTM type registry enabling config-only extension to new proteins and PTM types, (ii) EGFR/ERBB2 as a case study, and (iii) a second case study (ABL1/BCR-ABL, CML, 3 PTM types: phospho + acetylation + ubiquitination) demonstrating framework generality. This requires ~2 weeks of engineering without changing the core architecture or biological results.

7. Proposed Paper Structure

Section	Content Summary	Pages
Abstract	Problem, PTM-BDL approach, key results, biological discoveries	~250 words
Introduction	TKI resistance, PTM signaling, DRP limitations, contributions	1.5
Methods	Architecture (§2.1–2.2), data integration (§2.3), training, evaluation	3
Results	Performance, ablation, biological discoveries, benchmarking, LOCLO	3
Discussion	Interpretability contribution, comparison context, extensibility	1.5
Supplementary	Per-drug tables, CV details, full IG rankings, attention matrices	5+

Table 5. *Proposed paper structure.*

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Reference numbers correspond to PAPER_REFERENCES.md. Selected key references shown below.

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Full reference list (70 references) available in docs/PAPER_REFERENCES.md